

Introduction to Abstract Algebra III: Homework 4

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Exercise 43.4. Find all conjugates in $\mathbb{C}$ of $\sqrt{2} - \sqrt{3}$ over $\mathbb{Q}$.

Solution. Let $x = \sqrt{2} - \sqrt{3}$. Squaring both sides, we get:

$$x^2 = (\sqrt{2} - \sqrt{3})^2 = 2 - 2\sqrt{6} + 3 = 5 - 2\sqrt{6}.$$

Re-arranging, we get $-x^2 + 5 = 2\sqrt{6}$. Squaring both sides again, we get:

$$(x^2 - 5)^2 = 4 \cdot 6 = 24.$$

Expanding the left-hand side, we get $x^4 - 10x^2 + 25 = 24$. Re-arranging, we get $x^4 - 10x^2 + 1 = 0$.

We show that $f(x) = x^4 - 10x^2 + 1$ is a minimal polynomial of x over $\mathbb{Q}$. Since $f(x)$ is a monic polynomial with rational coefficients, it suffices to show that $f(x) = 0$ and that $f(x)$ is irreducible over $\mathbb{Q}$. Clearly, $f(x) = 0$ by construction.

We now show that $f(x)$ is irreducible over $\mathbb{Q}$. We start by showing that $f(x)$ has no linear factors over $\mathbb{Q}$. Using the Rational Root Test, the only possible rational roots of $f(x)$ are ± 1 . We have $f(1) = 1 - 10 + 1 = -8 \neq 0$ and $f(-1) = 1 - 10 + 1 = -8 \neq 0$. Thus, $f(x)$ has no linear factors over $\mathbb{Q}$.

Next, we show that $f(x)$ has no quadratic factors over $\mathbb{Q}$. If $f(x)$ has a quadratic factor over $\mathbb{Q}$, then it must be of the form $x^2 + ax + b$ for some $a, b \in \mathbb{Q}$. We have:

$$\begin{aligned} f(x) &= (x^2 + ax + b)(x^2 + cx + d) \\ &= x^4 + (a+c)x^3 + (ac+b+d)x^2 + (ad+bc)x + bd. \end{aligned}$$

Comparing coefficients, we get the following system of equations:

$$\begin{aligned} a + c &= 0, \\ ac + b + d &= -10, \\ ad + bc &= 0, \\ bd &= 1. \end{aligned}$$

From the first equation, we get $c = -a$. Substituting into the second equation, we get $-a^2 + b + d = -10$. Substituting into the third equation, we get $ad - ab = 0$, which implies that $d = b$. Substituting into the fourth equation, we get $b^2 = 1$, which implies that $b = \pm 1$. If $b = 1$, then we have $-a^2 + 1 + d = -10$, which implies that $d = -9 + a^2$. Substituting into the fourth equation, we get $1 \cdot (-9 + a^2) = 1$, which implies that $a^2 = 10$. This is not possible since $a \in \mathbb{Q}$. If $b = -1$, then we have $-a^2 - 1 + d = -10$, which implies that $d = -9 + a^2$. Substituting into the fourth equation, we get $(-1)(-9 + a^2) = 1$, which implies that $a^2 = 8$. This is not possible since $a \in \mathbb{Q}$. Thus, $f(x)$ has no quadratic factors over $\mathbb{Q}$.

Lastly, we show that $f(x)$ has no cubic factors over $\mathbb{Q}$. If $f(x)$ has a cubic factor over $\mathbb{Q}$, then it must be of the form $x^3 + ax^2 + bx + c$ for some $a, b, c \in \mathbb{Q}$. We have:

$$\begin{aligned} f(x) &= (x^3 + ax^2 + bx + c)(x + d) \\ &= x^4 + (a+d)x^3 + (b+ad)x^2 + (c+bd)x + cd. \end{aligned}$$

Comparing coefficients, we get the following system of equations:

$$\begin{aligned} a + d &= 0, \\ b + ad &= -10, \\ c + bd &= 0, \\ cd &= 1. \end{aligned}$$

From the first equation, we get $d = -a$. Substituting into the second equation, we get $b - a^2 = -10$. Substituting into the third equation, we get $c - ab = 0$, which implies that $c = ab$. Substituting into the

fourth equation, we get $ab(-a) = 1$, which implies that $a^2b = -1$. Substituting into the second equation, we get $b - a^2 = -10$, which implies that $b = -10 + a^2$. Substituting into the previous equation, we get $a^2(-10 + a^2) = -1$, which implies that $a^4 - 10a^2 + 1 = 0$. This is not possible since $a \in \mathbb{Q}$. Thus, $f(x)$ has no cubic factors over $\mathbb{Q}$.

Therefore, $f(x)$ is irreducible over $\mathbb{Q}$, making $f(x)$ the minimal polynomial of x over $\mathbb{Q}$.

We now determine the conjugates of α over $\mathbb{Q}$. Recall that the conjugates of α are the images $\sigma(\alpha)$ under field embeddings:

$$\sigma : \mathbb{Q}(\alpha) \hookrightarrow \mathbb{C} \quad \text{with} \quad \sigma|_{\mathbb{Q}} = \text{Id}.$$

Since $f(\alpha) = 0$ and σ fixes $\mathbb{Q}$, we have:

$$0 = \sigma(f(\alpha)) = f(\sigma(\alpha)),$$

so $\sigma(\alpha)$ must be a root of $f(x)$. Thus every conjugate of α is a root of its minimal polynomial.

Conversely, if β is any root of $f(x)$, then there exists an embedding:

$$\sigma : \mathbb{Q}(\alpha) \hookrightarrow \mathbb{C} \quad \text{such that} \quad \sigma(\alpha) = \beta.$$

Hence the conjugates of α are exactly the roots of $f(x)$.

It remains to compute the roots of $f(x)$. Observe that

$$x^4 - 10x^2 + 1 = 0 \iff x^2 = 5 \pm 2\sqrt{6}.$$

Therefore, the conjugates of $\alpha = \sqrt{2} - \sqrt{3}$ over $\mathbb{Q}$ are:

$$\sqrt{5 + 2\sqrt{6}}, \quad -\sqrt{5 + 2\sqrt{6}}, \quad \sqrt{5 - 2\sqrt{6}}, \quad -\sqrt{5 - 2\sqrt{6}},$$

which can be simplified to:

$$\sqrt{2} - \sqrt{3}, \quad -\sqrt{2} + \sqrt{3}, \quad \sqrt{2} + \sqrt{3}, \quad -\sqrt{2} - \sqrt{3}. \quad \square$$

Exercise 43.5. Find all conjugates in $\mathbb{C}$ of $\sqrt{2} + i$ over $\mathbb{Q}$.

Solution. We start by finding the minimal polynomial of $\alpha = \sqrt{2} + i$ over $\mathbb{Q}$. We have $x^2 = (\sqrt{2} + i)^2 = 1 + 2\sqrt{2}i$. Re-arranging, we get $x^2 - 1 = 2\sqrt{2}i$. Squaring both sides, we get $x^4 - 2x^2 + 1 = -8$, which can be re-arranged to $x^4 - 2x^2 + 9 = 0$. Using the same procedure as Exercise 43.4, we can show that $f(x)$ is irreducible over $\mathbb{Q}$, making $f(x)$ the minimal polynomial of α over $\mathbb{Q}$.

Again, using my work from Exercise 43.4, we can show that the conjugates of α over $\mathbb{Q}$ are exactly the roots of $f(x)$. We have:

$$x^4 - 2x^2 + 9 = 0 \iff x^2 = 1 \pm 2i\sqrt{2}.$$

Thus, the roots of $f(x)$ are:

$$\sqrt{1 + 2i\sqrt{2}}, \quad -\sqrt{1 + 2i\sqrt{2}}, \quad \sqrt{1 - 2i\sqrt{2}}, \quad -\sqrt{1 - 2i\sqrt{2}},$$

which can be simplified to:

$$\sqrt{2} + i, \quad -\sqrt{2} - i, \quad \sqrt{2} - i, \quad -\sqrt{2} + i. \quad \square$$

Exercise 43.6. Find all conjugates in $\mathbb{C}$ of $\sqrt{2} + i$ over $\mathbb{R}$.

Solution. We start by finding the vanishing. Since $\sqrt{2} \in \mathbb{R}$, we have $x - \sqrt{2} = i$. Squaring both sides, we get $(x - \sqrt{2})^2 = -1$, which can be re-arranged to $x^2 - 2\sqrt{2}x + 3 = 0$. Using the same procedure as Exercise 43.4, we can show that $f(x) = x^2 - 2\sqrt{2}x + 3$ is irreducible over $\mathbb{R}$, making $f(x)$ the minimal polynomial of α over $\mathbb{R}$.

Again, using my work from Exercise 43.4, we can show that the conjugates of α over $\mathbb{R}$ are exactly the roots of $f(x)$. We have:

$$x^2 - 2\sqrt{2}x + 3 = 0 \iff x = \sqrt{2} \pm i.$$

Thus, the conjugates of α over $\mathbb{R}$ are $\sqrt{2} + i$ and $\sqrt{2} - i$. $\square$

Exercise 43.7. Find all conjugates in $\mathbb{C}$ of $\sqrt{1 + \sqrt{2}}$ over $\mathbb{Q}$.

Solution. We start by finding the vanishing polynomial of $\alpha = \sqrt{1 + \sqrt{2}}$. Re-writing, we have $\alpha^2 - 1 = \sqrt{2}$. Squaring both sides, we get $\alpha^4 - 2\alpha^2 - 1 = 0$. Therefore, we have the vanishing polynomial $f(x) = x^4 - 2x^2 - 1$. Using the same procedure as Exercise 43.4, we can show that $f(x)$ is irreducible over $\mathbb{Q}$, making $f(x)$ the minimal polynomial of α over $\mathbb{Q}$.

Now, applying the same procedure as Exercise 43.4, we can show that the conjugates of α over $\mathbb{Q}$ are exactly the roots of $f(x)$. We have:

$$x^4 - 2x^2 - 1 = 0 \iff x^2 = 1 \pm \sqrt{2}.$$

Thus, the roots of $f(x)$ are:

$$\sqrt{1 + \sqrt{2}}, \quad -\sqrt{1 + \sqrt{2}}, \quad \sqrt{1 - \sqrt{2}}, \quad -\sqrt{1 - \sqrt{2}}. \quad \square$$

We consider the field $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. It can be shown that $[E : \mathbb{Q}] = 8$. In the notation of Theorem 43.18, we have the following conjugation isomorphisms (which are here automorphisms of E):

$$\begin{aligned} \psi_{\sqrt{2}, -\sqrt{2}} : (\mathbb{Q}(\sqrt{3}, \sqrt{5}))(\sqrt{2}) &\rightarrow (\mathbb{Q}(\sqrt{3}, \sqrt{5}))(-\sqrt{2}), \\ \psi_{\sqrt{3}, -\sqrt{3}} : (\mathbb{Q}(\sqrt{2}, \sqrt{5}))(\sqrt{3}) &\rightarrow (\mathbb{Q}(\sqrt{2}, \sqrt{5}))(-\sqrt{3}), \\ \psi_{\sqrt{5}, -\sqrt{5}} : (\mathbb{Q}(\sqrt{2}, \sqrt{3}))(\sqrt{5}) &\rightarrow (\mathbb{Q}(\sqrt{2}, \sqrt{3}))(-\sqrt{5}). \end{aligned}$$

For shorter notation, let $\tau_2 = \psi_{\sqrt{2}, -\sqrt{2}}$ and $\tau_3 = \psi_{\sqrt{3}, -\sqrt{3}}$ and $\tau_5 = \psi_{\sqrt{5}, -\sqrt{5}}$.

Exercise 43.10. Compute $\tau_2(\sqrt{2} + \sqrt{5})$.

Solution. Computing $\tau_2(\sqrt{2} + \sqrt{5})$, we have:

$$\begin{aligned} \tau_2(\sqrt{2} + \sqrt{5}) &= \tau_2(\sqrt{2}) + \tau_2(\sqrt{5}) \\ &= -\sqrt{2} + \sqrt{5}. \end{aligned} \quad \square$$

Exercise 43.12. Compute $(\tau_5\tau_3)\left(\frac{\sqrt{2}-3\sqrt{5}}{2\sqrt{3}-\sqrt{2}}\right)$.

Solution. Computing $\tau_5\tau_3\left(\frac{\sqrt{2}-3\sqrt{5}}{2\sqrt{3}-\sqrt{2}}\right)$, we have:

$$\begin{aligned} \tau_5\tau_3\left(\frac{\sqrt{2}-3\sqrt{5}}{2\sqrt{3}-\sqrt{2}}\right) &= \tau_5\left(\frac{\tau_3(\sqrt{2})-3\tau_3(\sqrt{5})}{2\tau_3(\sqrt{3})-\tau_3(\sqrt{2})}\right) \\ &= \tau_5\left(\frac{\sqrt{2}-3\sqrt{5}}{-2\sqrt{3}-\sqrt{2}}\right) \\ &= \frac{\tau_5(\sqrt{2})-3\tau_5(\sqrt{5})}{-2\tau_5(\sqrt{3})-\tau_5(\sqrt{2})} \\ &= \frac{\sqrt{2}+3\sqrt{5}}{-2\sqrt{3}-\sqrt{2}}. \end{aligned} \quad \square$$

Exercise 43.15. Compute $\tau_3\tau_5\tau_3^{-1}(\sqrt{3}-\sqrt{5})$.

Solution. Computing $\tau_3\tau_5\tau_3^{-1}(\sqrt{3}-\sqrt{5})$, we have:

$$\begin{aligned}
 \tau_3\tau_5\tau_3^{-1}(\sqrt{3}-\sqrt{5}) &= \tau_3\tau_5\left(\frac{\sqrt{3}-\sqrt{5}}{\tau_3(\sqrt{3}-\sqrt{5})}\right) \\
 &= \tau_3\tau_5\left(\frac{\sqrt{3}-\sqrt{5}}{-\sqrt{3}-\sqrt{5}}\right) \\
 &= \tau_3\left(\frac{\tau_5(\sqrt{3})-\tau_5(\sqrt{5})}{-\tau_5(\sqrt{3})-\tau_5(\sqrt{5})}\right) \\
 &= \tau_3\left(\frac{\sqrt{3}+\sqrt{5}}{-\sqrt{3}+\sqrt{5}}\right) \\
 &= \frac{\tau_3(\sqrt{3})+\tau_3(\sqrt{5})}{-\tau_3(\sqrt{3})+\tau_3(\sqrt{5})} \\
 &= \frac{-\sqrt{3}+\sqrt{5}}{\sqrt{3}+\sqrt{5}}.
 \end{aligned}$$

□

Exercise 43.18. Find the fixed field of the automorphism or set of automorphisms of E , $\{\tau_2, \tau_3\}$.

Solution. The automorphisms act on the basis elements of E by changing signs. Specifically, τ_2 maps $\sqrt{2} \mapsto -\sqrt{2}$ and fixes $\sqrt{3}, \sqrt{5}$ and τ_3 maps $\sqrt{3} \mapsto -\sqrt{3}$ and fixes $\sqrt{2}, \sqrt{5}$. An element $\alpha \in E$ is written as:

$$\alpha = a_0 + a_1\sqrt{2} + a_2\sqrt{3} + a_3\sqrt{5} + a_4\sqrt{6} + a_5\sqrt{10} + a_6\sqrt{15} + a_7\sqrt{30}.$$

For $\tau_2(\alpha) = \alpha$, all terms involving $\sqrt{2}$ (i.e., $\sqrt{2}, \sqrt{6}, \sqrt{10}, \sqrt{30}$) must have coefficients equal to zero. This simplifies α to:

$$\alpha = a_0 + a_2\sqrt{3} + a_3\sqrt{5} + a_6\sqrt{15}.$$

For $\tau_3(\alpha) = \alpha$ on the remaining terms, all terms involving $\sqrt{3}$ (i.e., $\sqrt{3}, \sqrt{15}$) must have coefficients equal to zero. This leaves:

$$\alpha = a_0 + a_3\sqrt{5}.$$

Thus, the fixed field is $\{a_0 + a_3\sqrt{5} \mid a_0, a_3 \in \mathbb{Q}\}$, which is exactly $\mathbb{Q}(\sqrt{5})$. □

Exercise 43.20. Find the fixed field of the automorphism or set of automorphisms of E , $\tau_5\tau_3\tau_2$.

Solution. Notice that the basis of E is all products of $\sqrt{2}, \sqrt{3}, \sqrt{5}$, so it suffices to compute $\tau_5\tau_3\tau_2$ on these basis elements. We have:

$$\begin{aligned}
 \sigma(\sqrt{6}) &= \sigma(\sqrt{2})\sigma(\sqrt{3}) = -\sqrt{2} \cdot -\sqrt{3} = \sqrt{6} \\
 \sigma(\sqrt{10}) &= \sigma(\sqrt{2})\sigma(\sqrt{5}) = \sqrt{10} \\
 \sigma(\sqrt{15}) &= \sigma(\sqrt{3})\sigma(\sqrt{5}) = \sqrt{15} \\
 \sigma(\sqrt{30}) &= \sigma(\sqrt{2})\sigma(\sqrt{3})\sigma(\sqrt{5}) = -\sqrt{30}.
 \end{aligned}$$

Thus, for any $\alpha \in E$, we have $\sigma(\alpha) = -a - b\sqrt{2} - c\sqrt{3} - d\sqrt{5} + e\sqrt{6} + f\sqrt{10} + g\sqrt{15} - h\sqrt{30}$. So $\alpha = \sigma(\alpha)$ if and only if

$$\begin{aligned}
 a &= -a = 0 \\
 b &= -b = 0 \\
 c &= -c = 0 \\
 d &= -d = 0
 \end{aligned}$$

$$\begin{aligned}
e &= e \\
f &= f \\
g &= g \\
h &= -h = 0.
\end{aligned}$$

So the elements $\alpha \in E$ fixed by σ are of the form $e\sqrt{6} + f\sqrt{10} + g\sqrt{15}$ which are exactly the elements of $\mathbb{Q}(\sqrt{6}, \sqrt{10}, \sqrt{15}) = \mathbb{Q}(\sqrt{6}, \sqrt{10})$. Thus $E_{\tau_5\tau_3\tau_2} = \mathbb{Q}(\sqrt{6}, \sqrt{10})$. $\square$

Exercise 43.22.

- (i) Show that each of the automorphisms τ_2 , τ_3 , and τ_5 is of order 2 in $G(E/\mathbb{Q})$. (Remember what is meant by the order of an element of a group.)
- (ii) Find the subgroup H of $G(E/\mathbb{Q})$ generated by the elements τ_2 , τ_3 , and τ_5 , and give the group table. [Hint: There are eight elements.]
- (iii) Just as was done in Example 43.4, argue that the group H of part (b) is the full group $G(E/\mathbb{Q})$.

Solution to (i). For τ_2 , we have $\tau_2(\sqrt{2}) = -\sqrt{2}$ and τ_2 fixes $\sqrt{3}$ and $\sqrt{5}$. Applying it twice, we get $\tau_2^2(\sqrt{2}) = -(-\sqrt{2}) = \sqrt{2}$. Since τ_2^2 fixes the generators $\sqrt{2}, \sqrt{3}, \sqrt{5}$, it fixes every element in E . Thus $\tau_2^2 = \iota$. Since $\tau_2 \neq \iota$, the order is 2. The same logic applies to τ_3 and τ_5 . $\square$

Solution to (ii). The subgroup $H = \langle \tau_2, \tau_3, \tau_5 \rangle$ consists of all compositions of these maps. Since they commute (e.g., $\tau_2\tau_3 = \tau_3\tau_2$) and each has order 2, the elements are:

$$H = \{\iota, \tau_2, \tau_3, \tau_5, \tau_2\tau_3, \tau_2\tau_5, \tau_3\tau_5, \tau_2\tau_3\tau_5\}.$$

Letting $\tau_{ijk} = \tau_i\tau_j \dots$, the group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

$\circ$	ι	τ_2	τ_3	τ_5	τ_{23}	τ_{25}	τ_{35}	τ_{235}
ι	ι	τ_2	τ_3	τ_5	τ_{23}	τ_{25}	τ_{35}	τ_{235}
τ_2	τ_2	ι	τ_{23}	τ_{25}	τ_3	τ_5	τ_{235}	τ_{35}
τ_3	τ_3	τ_{23}	ι	τ_{35}	τ_2	τ_{235}	τ_5	τ_{25}
τ_5	τ_5	τ_{25}	τ_{35}	ι	τ_{235}	τ_2	τ_3	τ_{23}
τ_{23}	τ_{23}	τ_3	τ_2	τ_{235}	ι	τ_{35}	τ_{25}	τ_5
τ_{25}	τ_{25}	τ_5	τ_{235}	τ_2	τ_{35}	ι	τ_{23}	τ_3
τ_{35}	τ_{35}	τ_{235}	τ_5	τ_3	τ_{25}	τ_{23}	ι	τ_2
τ_{235}	τ_{235}	τ_{35}	τ_{25}	τ_{23}	τ_5	τ_3	τ_2	ι

The group table is symmetric since the group is abelian. $\square$

Solution to (iii). Any automorphism in $G(E/\mathbb{Q})$ must send $\sqrt{2}$, $\sqrt{3}$, and $\sqrt{5}$ to two possible values (i.e., $\sqrt{2} \mapsto \pm\sqrt{2}$, $\sqrt{3} \mapsto \pm\sqrt{3}$, and $\sqrt{5} \mapsto \pm\sqrt{5}$). Since there are 8 possible combinations of signs, there are at most 8 automorphisms in $G(E/\mathbb{Q})$. Since H has 8 elements, we must have $H = G(E/\mathbb{Q})$. $\square$